Announcement

We will have Quiz #2 on Tuesday, April 14.

Open book and open notes

Final Project Presentation Schedule

April 16	April 21	April 23
James (11+2)	Ty & JC (13+2)	Joseph (11+2)
Boshi (11+2)	Thomas (11+2)	Sofia (11+2)
Supriya (11+2)	Devon & Jean-Luc (13+2)	Cade (11+2)
Amirreza (11+2)	Vigneswaran (11+2)	Bharat (11+2)
		Nitin (11+2)

Please submit your presentation slides to Blackboard before your presentation.

Course Evaluation

The course evaluation is available in Blackboard.

Please complete online course evaluation at Blackboard, which closes in April 27

Your feedback is crucial to improving the course

Today's Agenda

Region-based image segmentation

Morphological image processing techniques

Image Segmentation

(a) $\bigcup_{i=1}^{n} R_i = R$

(b) R_i is a connected set, $i = 1,\ldots,n$

(c) $R_i \cap R_j = \phi, \forall i \neq j$

(d) $Q(R_i) = TRUE$

(e) $Q(R_i \cup R_j) = FALSE$ for adjacent regions R_i and R_j

Two categories based on intensity properties:

- Discontinuity – edge-based algorithms
- Similarity – region-based algorithms

Region-Based Segmentation

- Region growing
- Region splitting and merging

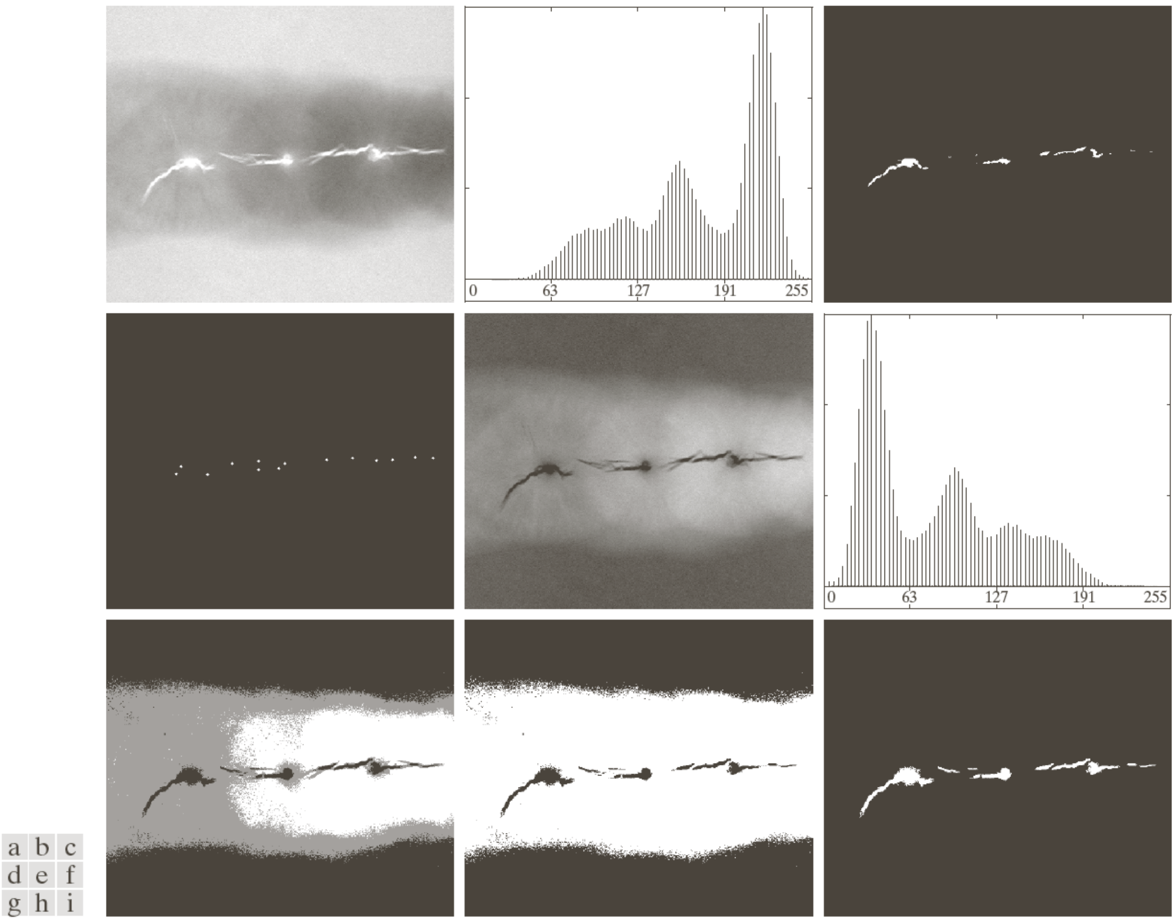
Region Growing Algorithm

-A procedure that groups pixels or subregions into larger regions based on predefined criteria for growth

-Start with a set of “seed” points and grow regions by appending neighboring pixels that satisfy the given criteria

- Connectivity
- Stopping rules
  - Local criteria: intensity values, textures, color
  - Prior knowledge: size and shape of the object

An Example



$$Q = \begin{cases} True & if\ |I_{seed} - I(x,y)| \le T \\ False & otherwise \end{cases}$$

FIGURE 10.51 (a) X-ray image of a defective weld. (b) Histogram. (c) Initial seed image. (d) Final seed image (the points were enlarged for clarity). (e) Absolute value of the difference between (a) and (c). (f) Histogram of (e). (g) Difference image thresholded using dual thresholds. (h) Difference image thresholded with the smallest of the dual thresholds. (i) Segmentation result obtained by region growing. (Original image courtesy of X-TEK Systems, Ltd.)

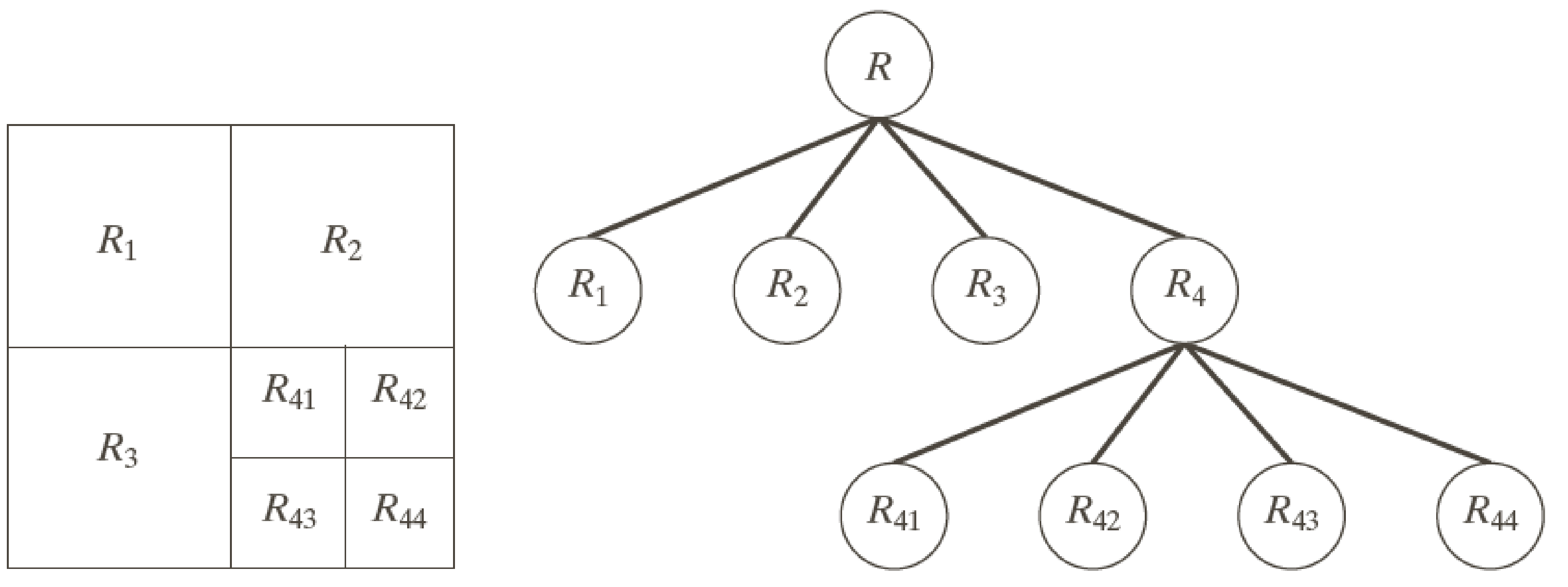
Region-Based Segmentation

- Region growing
- Region splitting and merging

Region-Splitting and Merging Algorithm

Step1: Keep splitting the region while $Q(R_i) = FALSE$ and $R_i > \min Size$

Step 2: Merge the subregions while $Q(R_i \cup R_j) = TRUE$



a b

FIGURE 10.52 (a) Partitioned image. (b) Corresponding quadtree. R represents the entire image region.

An Example

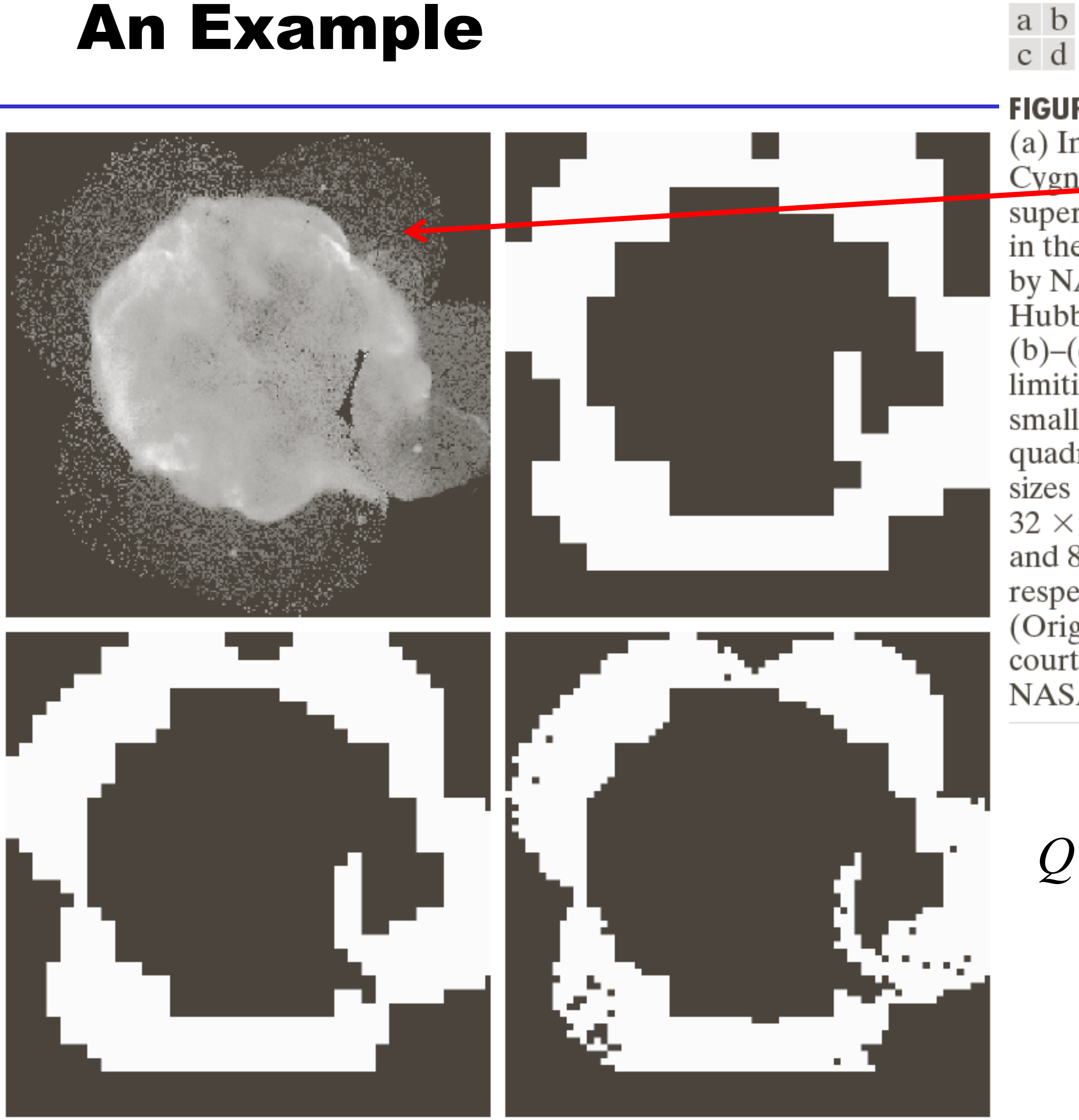


FIGURE 10.53 (a) Image of the Cygnus Loop supernova, taken in the X-ray band by NASA's Hubble Telescope. (b)–(d) Results of limiting the smallest allowed quadregion to sizes of 32 × 32, 16 × 16, and 8 × 8 pixels, respectively. (Original image courtesy of NASA.)

Extract the outer ring

$$Q = \begin{cases} TRUE & if \sigma > a \, \& \, 0 < m < b \\ FALSE & otherwise \end{cases}$$

Advanced Approaches for Image Segmentation

Image segmentation is still a research problem that is being investigated by many researchers

General-purpose image segmentation is far from well solved

- Image segmentation by K-means clustering
- Image segmentation with Graphic Models (MRF, CRF, etc.)

Semantic Segmentation with Deep Learning

http://blog.qure.ai/notes/semantic-segmentation-deep-learning-review

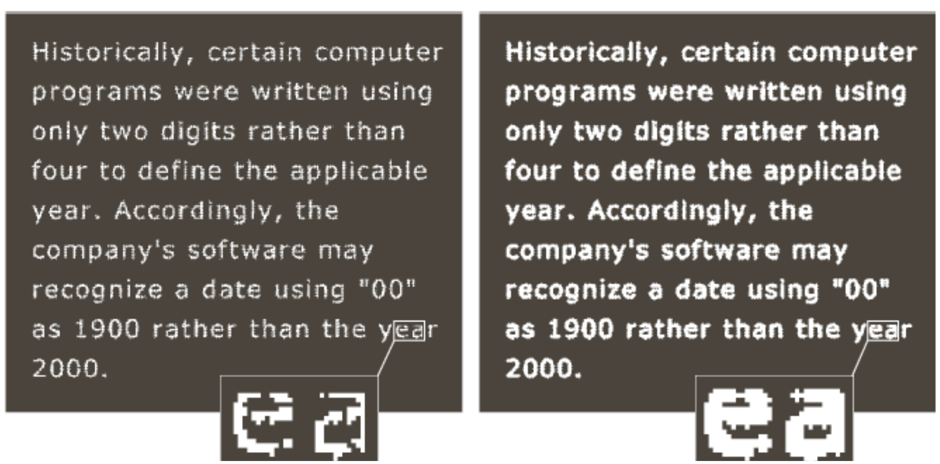
Morphological Image Processing – Techniques to Improve Image Segmentation

Objective: Extract image components for representation and description of region shape including

- Boundaries
- Skeletons
- Convex hull

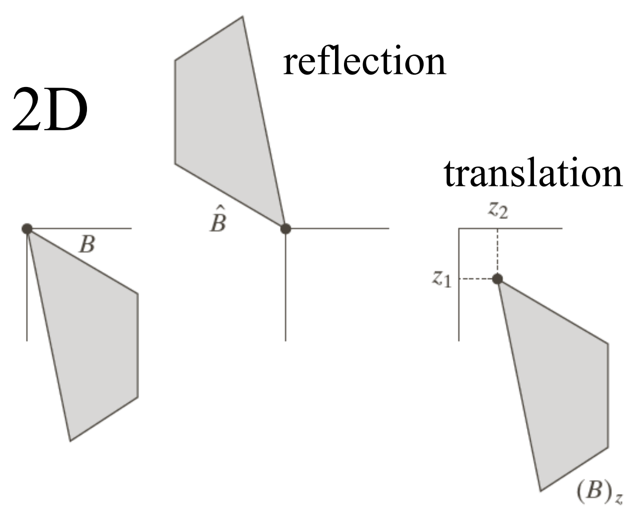
Applications:

- Edge detection
- Blob/connected component detection

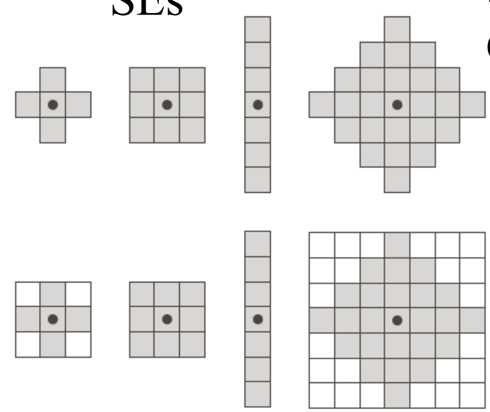


Basic Concepts

- **2D Integer space Z^2**
- **Union, intersection, complement, difference**
- **Set reflection** $\widehat{\mathcal{B}} = \{\mathbf{w} | \mathbf{w} = -\mathbf{b}, \mathbf{b} \in \mathcal{B}\}$ $\mathcal{B}$ is a set of 2D points (x, y)
- **Set translation** $(\mathcal{B})_z = \{\mathbf{c} | \mathbf{c} = \mathbf{b} + \mathbf{z}, \mathbf{b} \in \mathcal{B}\}$ -- move the center/origin of $\mathcal{B}$ by **z**
- **Structure elements (SEs): small sets/subimages used in morphology**



SEs



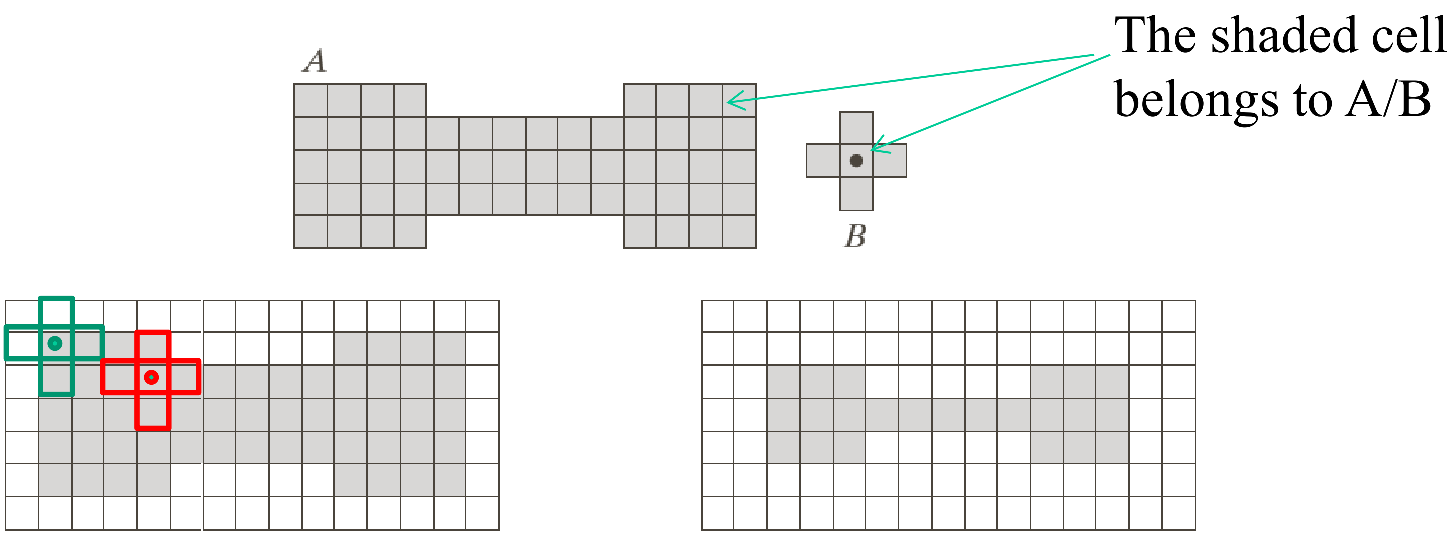
SEs for images

Black dots are centers/origins of SE

An Example of Morphology

Morphology - Create a new set by running B over A so that the origin of B visits every element of A.

An example of erosion: If B is completely contained in A for each operation, the new element is a member of the new set.



Erosion result

Common Morphological Operations

Two basic operations

- Erosion
- Dilation

Other operations

- Opening/closing
- Hit-or-Miss transform
- Thinning/thickening
- Hole filling

Erosion

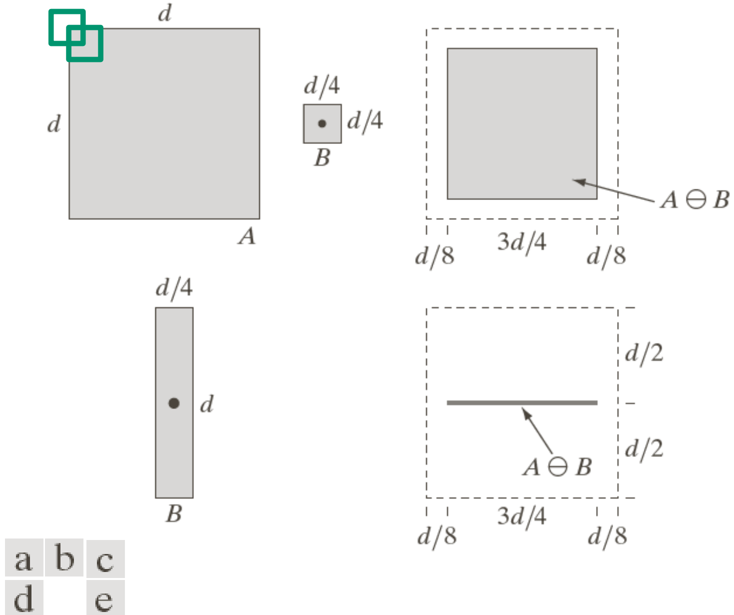


FIGURE 9.4 (a) Set A. (b) Square structuring element, B. (c) Erosion of A by B, shown shaded. (d) Elongated structuring element. (e) Erosion of A by B using this element. The dotted border in (c) and (e) is the boundary of set A, shown only for reference.

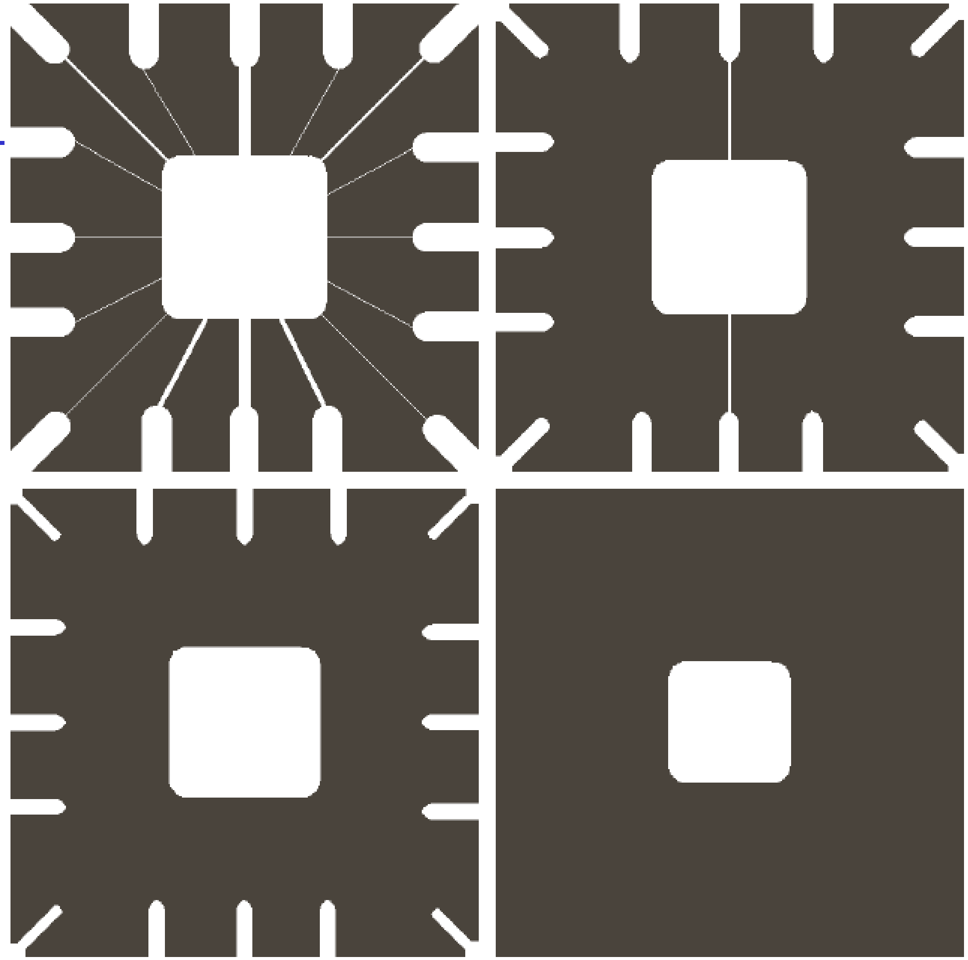
Set translation

$$(B)_z = \{\mathbf{c} \mid \mathbf{c} = \mathbf{b} + \mathbf{z}, \mathbf{b} \in B\}$$

$$A \ominus B = \{z \mid (B)_z \subseteq A\} \quad \text{or} \quad A \ominus B = \{z \mid (B)_z \cap A^c = \emptyset\}$$

Shrink or thin objects and remove the details smaller than the SE

Erosion



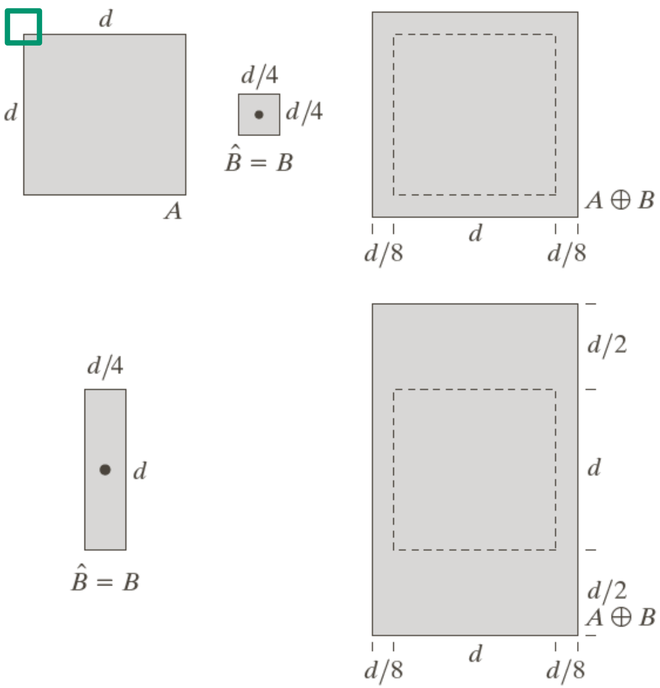
a b
c d

FIGURE 9.5 Using erosion to remove image components. (a) A 486 × 486 binary image of a wire-bond mask. (b)–(d) Image eroded using square structuring elements of sizes 11 × 11, 15 × 15, and 45 × 45, respectively. The elements of the SEs were all 1s.

$$A \ominus B = \{z \mid (B)_z \subseteq A\} \quad \text{or} \quad A \ominus B = \{z \mid (B)_z \cap A^c = \emptyset\}$$

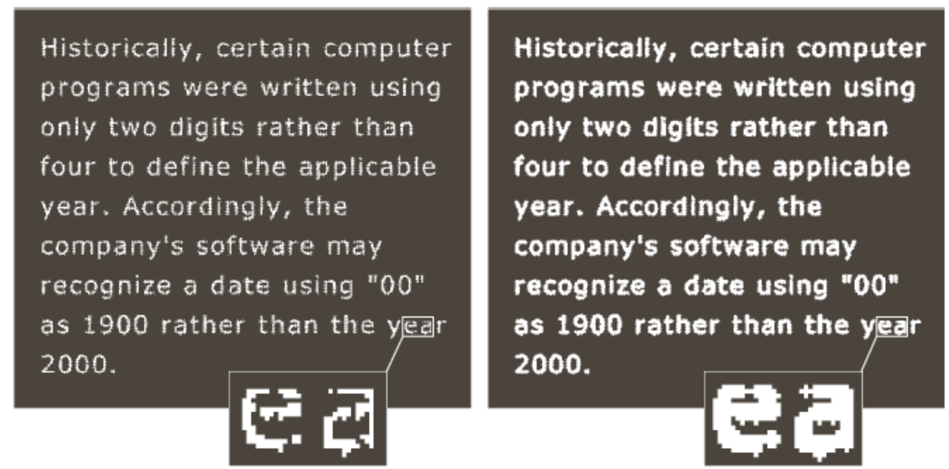
Shrink or thin objects and remove the details smaller than the SE

Dilation



a b c
d e

FIGURE 9.6
(a) Set A.
(b) Square structuring element (the dot denotes the origin).
(c) Dilation of A by B, shown shaded.
(d) Elongated structuring element. (e) Dilation of A using this element. The dotted border in (c) and (e) is the boundary of set A, shown only for reference



0	1	0
1	1	1
0	1	0

Set reflection about its origin: $\hat{B} = \{ w \mid w = -b, b \in B \}$

$$A \oplus B = \left\{ z \mid (\hat{B})_z \cap A \neq 0 \right\} \quad \text{or} \quad A \oplus B = \left\{ z \mid (\hat{B})_z \cap A \subseteq A \right\}$$

Grows or thickens objects and remove the gaps smaller than the SE

Properties of Dilation

- **Dilation is commutative** $A \oplus B = B \oplus A$

- **Dilation is associative** $A \oplus B \oplus C = A \oplus \left(B \oplus C\right)$

- **Dilation is distributive over the union operation**

$$A \oplus (B \cup C) = (A \oplus B) \cup (A \oplus C)$$

Properties of Erosion and Dilation

- **Erosion** $\boldsymbol{A} \ominus \boldsymbol{B} \ominus \mathbf{C} = \mathbf{A} \ominus (\boldsymbol{B} \oplus \boldsymbol{C})$

- **Erosion and dilation are duals of each other**

$$A \oplus B = \left(A^c \ominus \hat{B}\right)^c$$

Properties of Erosion and Dilation

- $A \subseteq (C \ominus B)$ **if and only if** $(A \oplus B) \subseteq C$

- **If** $A \subseteq C$, $A \oplus B \subseteq C \oplus B$ **and** $A \ominus B \subseteq C \ominus B$

Opening

- Smooth the contour of an object
- Break narrow bridges
- Eliminate thin protrusions

$$A \circ B = (A \ominus B) \oplus B$$

$$(A \circ B) \circ B = A \circ B$$

$$(A \circ B) \subseteq A$$

$$if\ A \subseteq C, A \circ B \subseteq C \circ B$$

$$A \circ B = \cup \{(B)_z | (B)_z \subseteq A\}$$

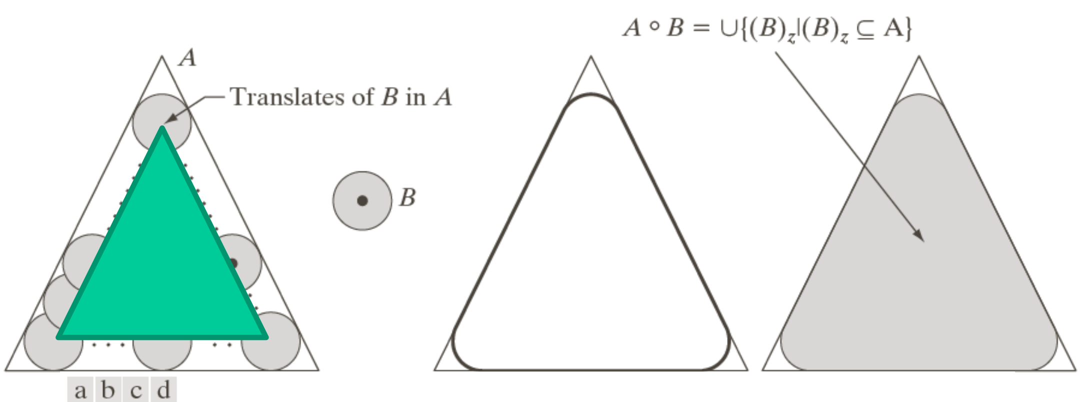
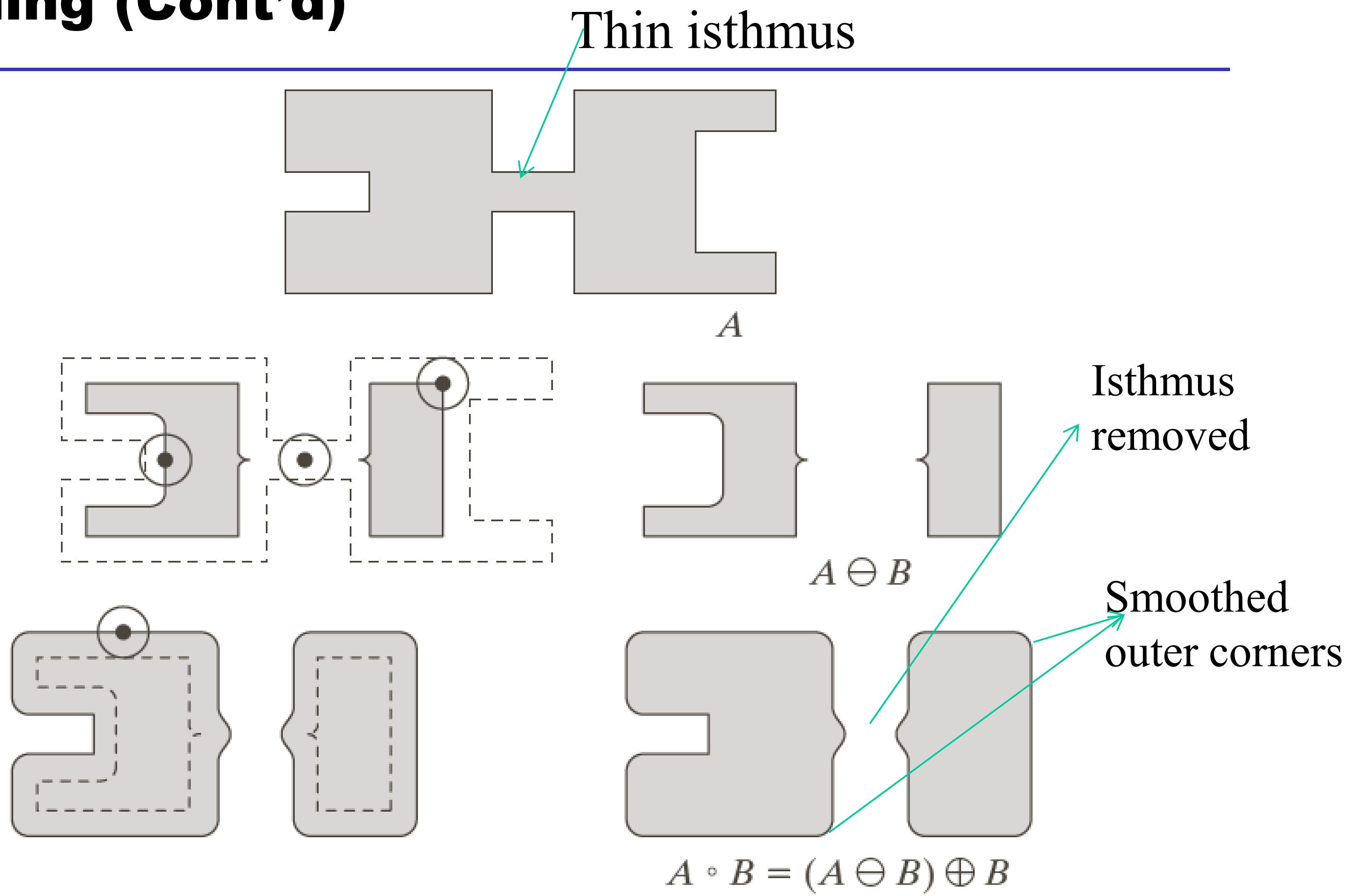


FIGURE 9.8 (a) Structuring element B "rolling" along the inner boundary of A (the dot indicates the origin of B). (b) Structuring element. (c) The heavy line is the outer boundary of the opening. (d) Complete opening (shaded). We did not shade A in (a) for clarity.

The SE rolls within the boundary of A.

Opening (Cont'd)



Closing

- **Smooth the contour of an object**
- **Fill narrow breaks and gaps**
- **Eliminate long and thin gulfs**
- **Eliminate small holes**

$$A \bullet B = (A \oplus B) \ominus B$$

$$(A \bullet B) \bullet B = A \bullet B$$

$$A \subseteq (A \bullet B)$$

$$if\, A \subseteq C, A \bullet B \subseteq C \bullet B$$

Opening and closing are duals of each other $\quad A \bullet B = (A^c \circ \hat{B})^c$

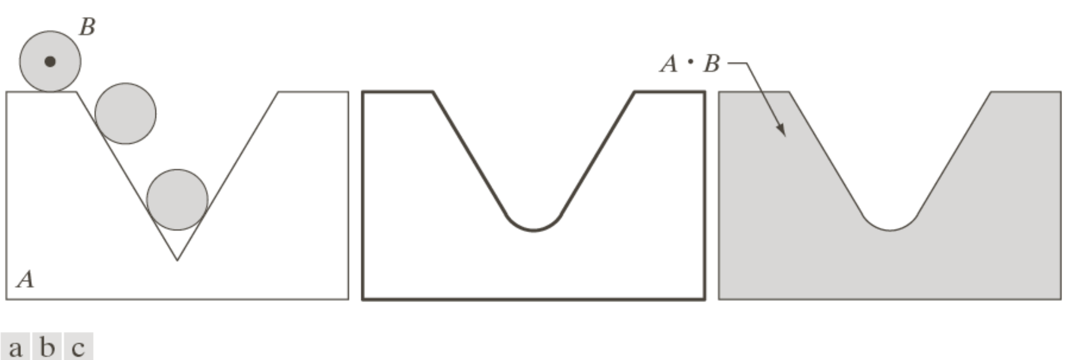
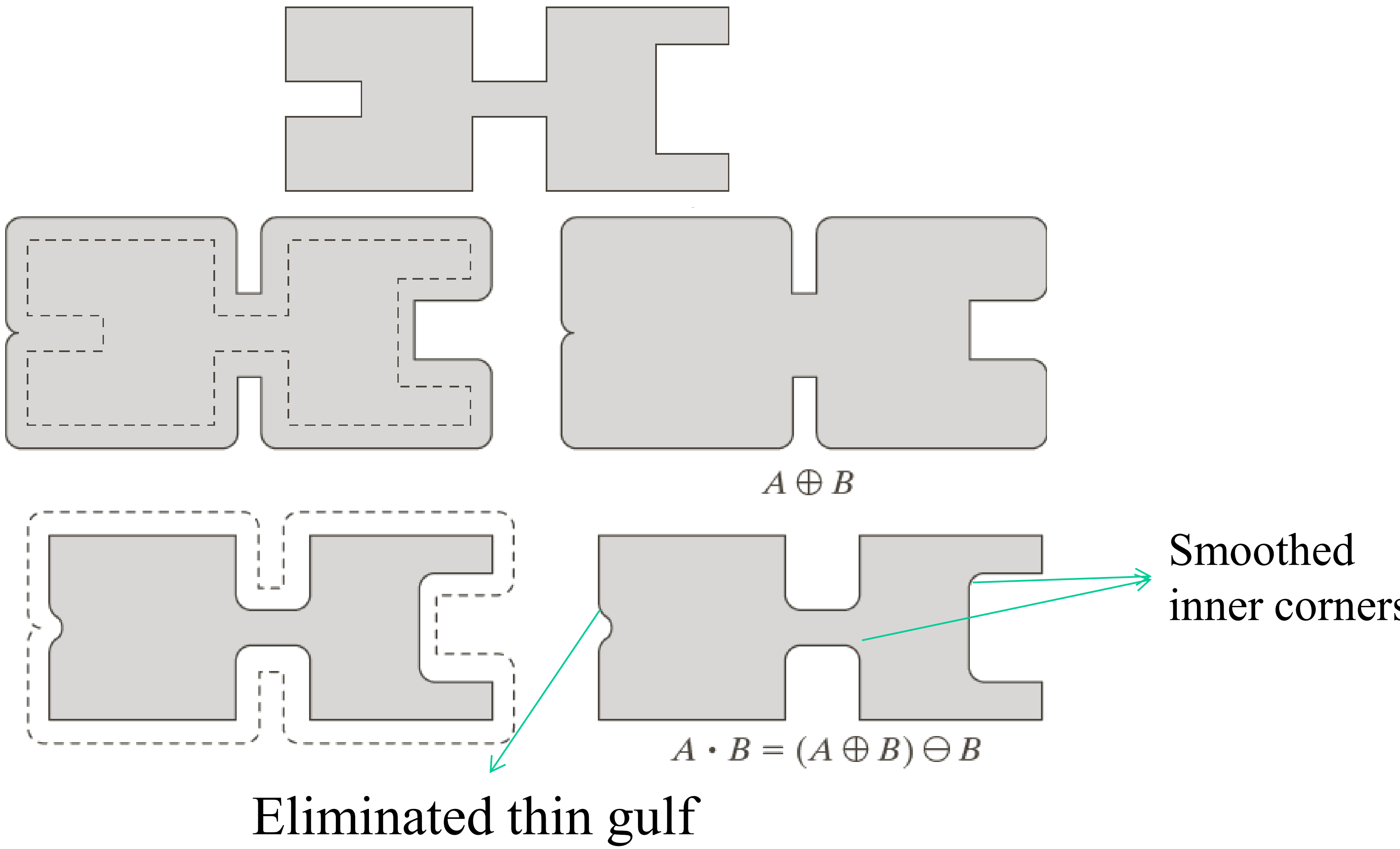


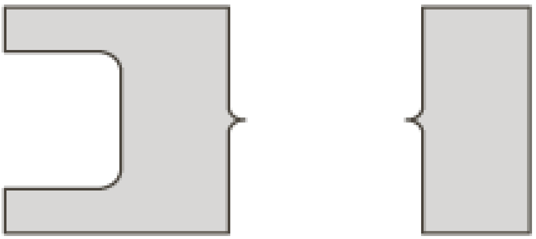
FIGURE 9.9 (a) Structuring element B "rolling" on the outer boundary of set A. (b) The heavy line is the outer boundary of the closing. (c) Complete closing (shaded). We did not shade A in (a) for clarity.

The SE rolls outside the boundary of A.

Closing (Cont'd)

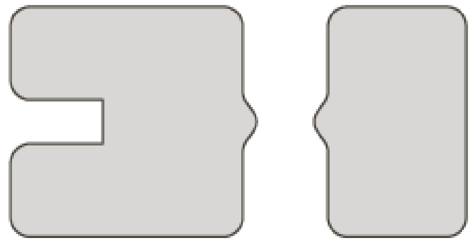


Opening & Closing



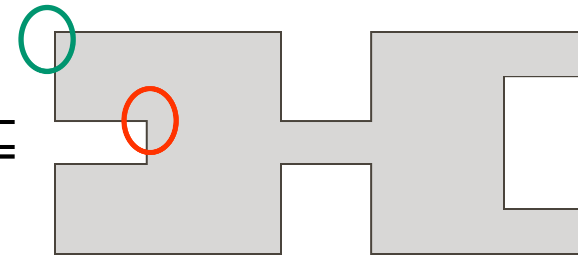
$A \ominus B$

$\subseteq$



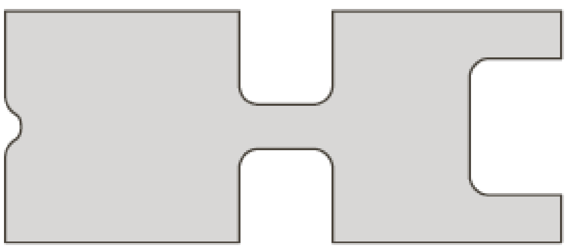
$A \circ B = (A \ominus B) \oplus B$

$\subseteq$



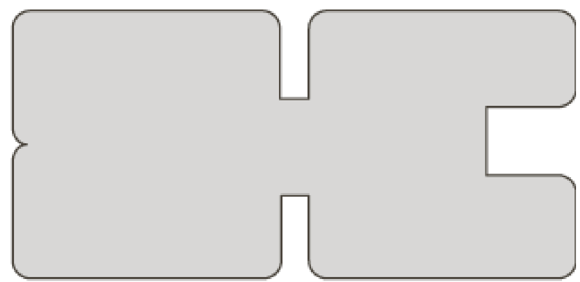
A

$\subseteq$



$A \bullet B = (A \oplus B) \ominus B$

$\subseteq$



$A \oplus B$

$$A \ominus B \subseteq A \circ B \subseteq A \subseteq A \bullet B \subseteq A \oplus B$$

An Example of Opening & Closing

- An opening removes all noise
    - removing the white noise by erosion
    - removing the black noise by dilation
- An additional closing fills the gaps

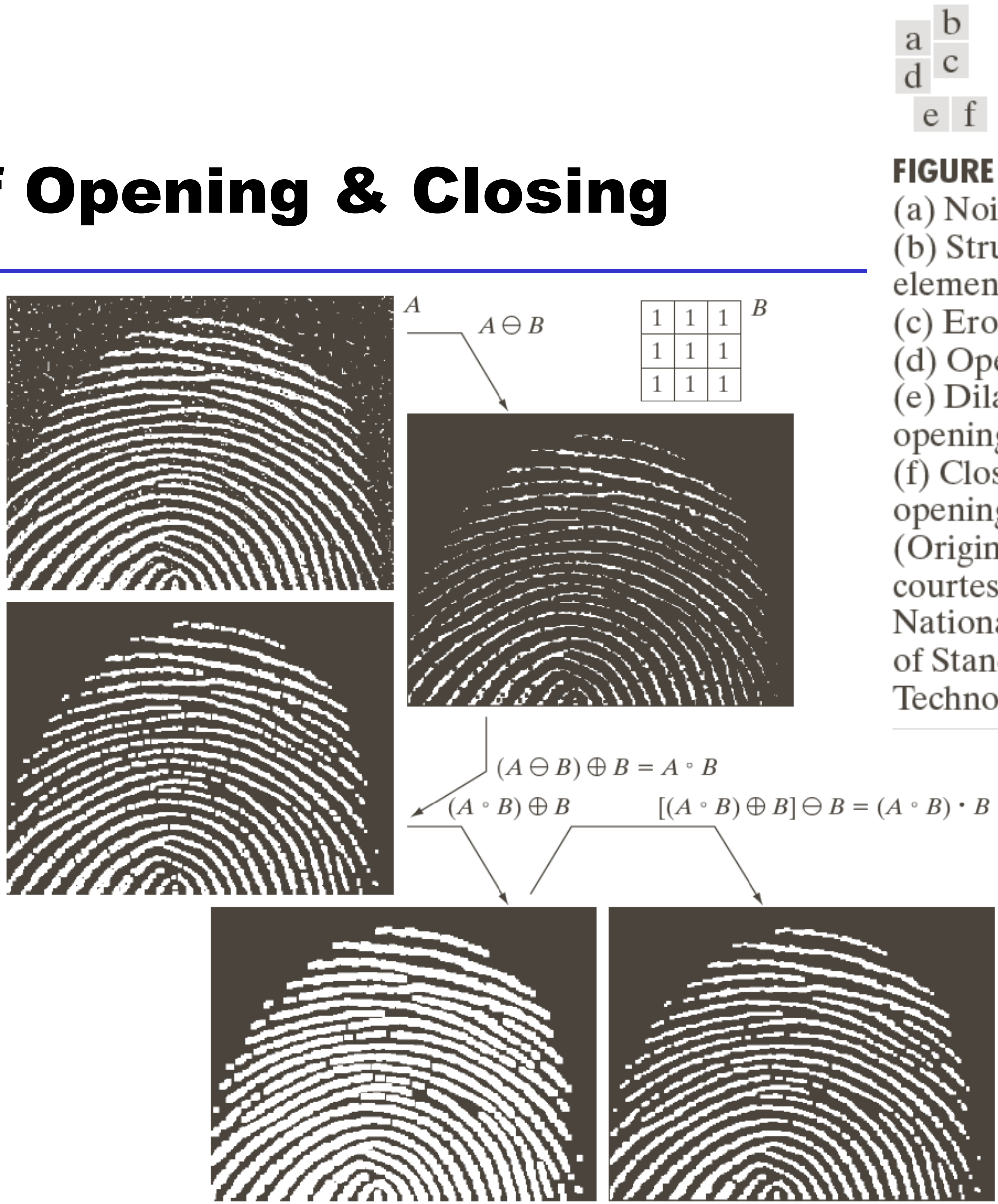


FIGURE 9.11 (a) Noisy image. (b) Structuring element. (c) Eroded image. (d) Opening of A. (e) Dilation of the opening. (f) Closing of the opening. (Original image courtesy of the National Institute of Standards and Technology.)

Hit-or-Miss Transform

Shape detection: find D

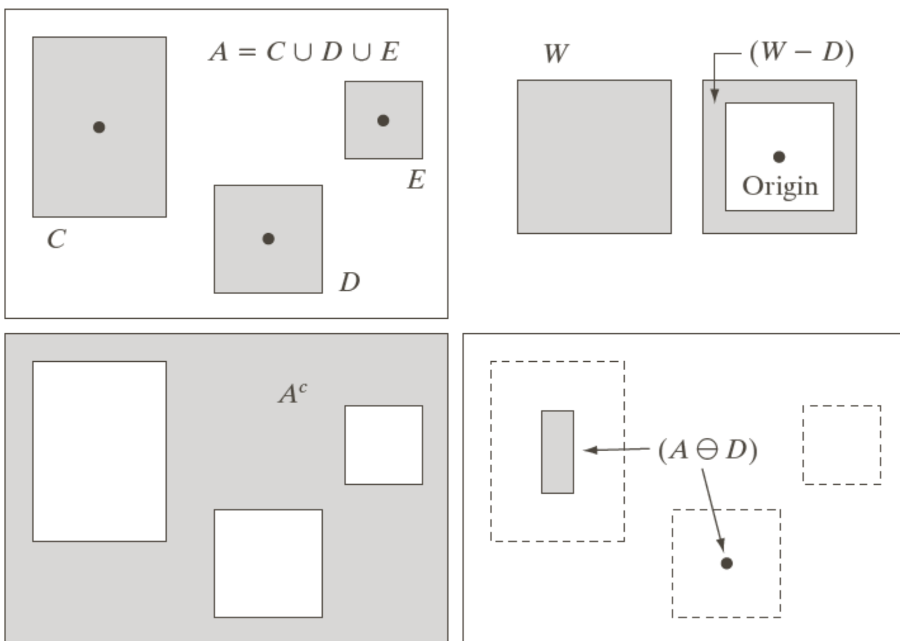
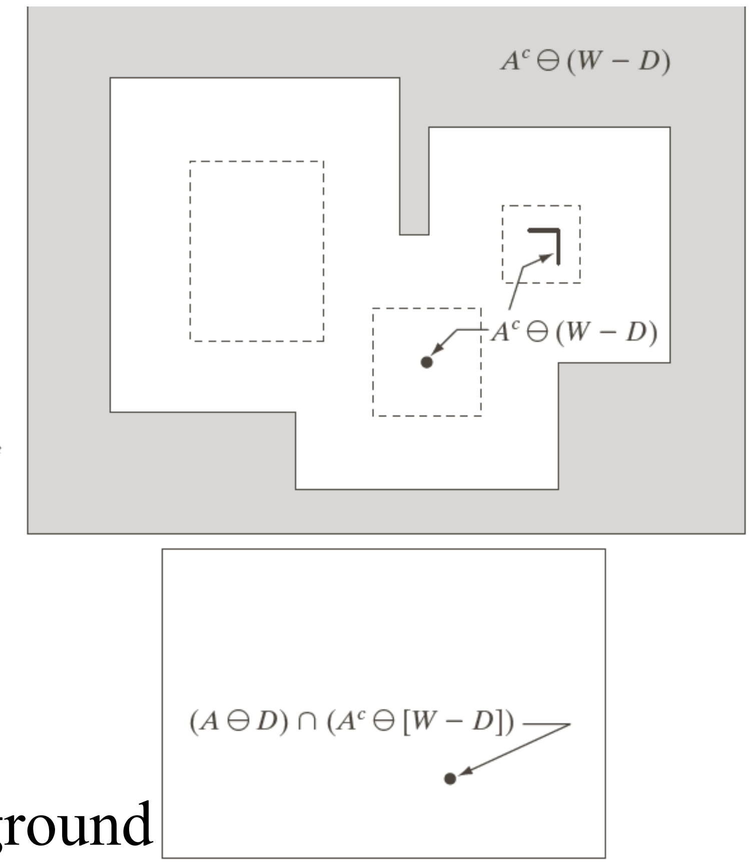


FIGURE 9.12 (a) Set A. (b) A window, W, and the local background of D with respect to W, $(W - D)$. (c) Complement of A. (d) Erosion of A by D. (e) Erosion of A^c by $(W - D)$. (f) Intersection of (d) and (e), showing the location of the origin of D, as desired. The dots indicate the origins of C, D, and E.

a	b	e
c	d	f



Define a set B consisting of D and its background

$$B = (D, W - D) \quad \Rightarrow \quad A \circledast B = (A \ominus D) \cap [A^c \ominus (W - D)]$$

Other Basic Morphological Algorithms

Thinning $A \otimes B = A - (A \circledast B)$

$$B = \{B^1, B^2, B^3, \cdots, B^n\} \Rightarrow A \otimes B = \left(\left(\left((A \otimes B^1) \otimes B^2\right) \cdots\right) \otimes B^n\right)$$

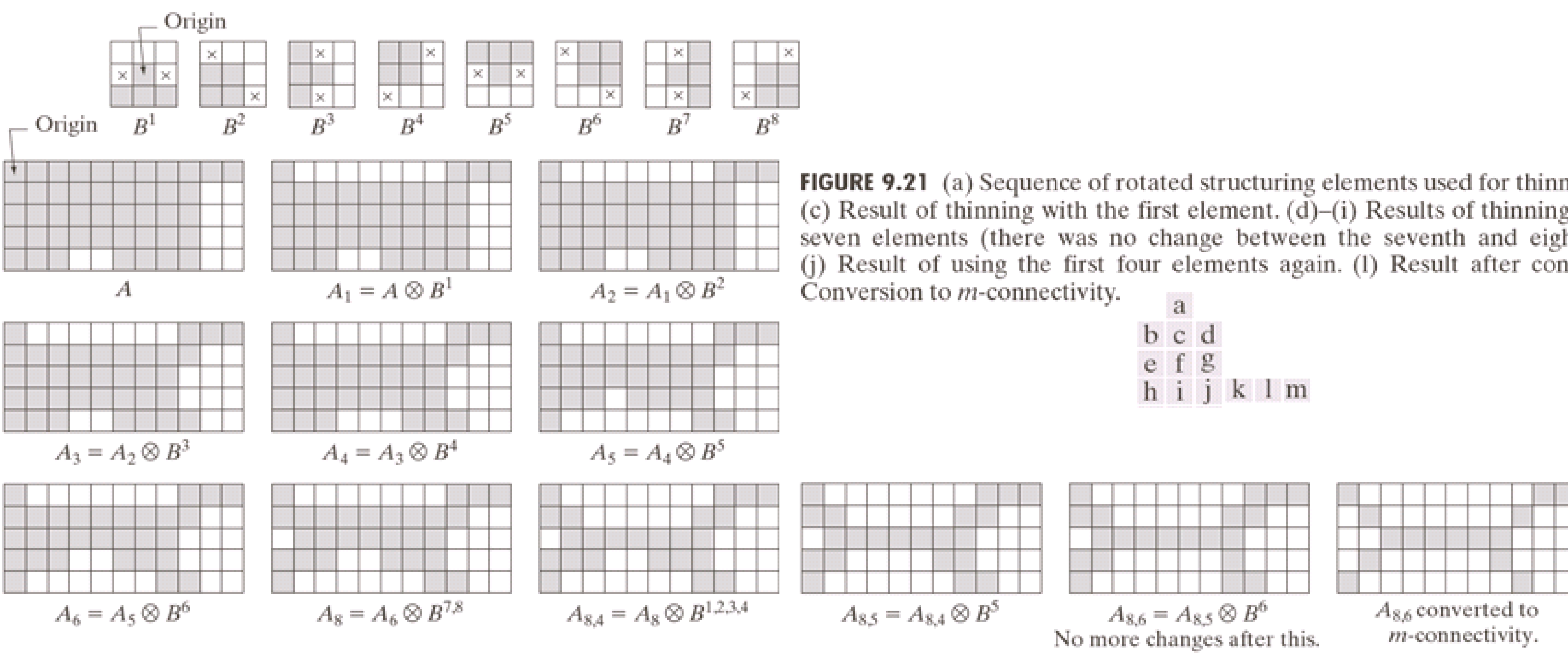


FIGURE 9.21 (a) Sequence of rotated structuring elements used for thinning. (b) Set A. (c) Result of thinning with the first element. (d)–(i) Results of thinning with the next seven elements (there was no change between the seventh and eighth elements). (j) Result of using the first four elements again. (l) Result after convergence. (m) Conversion to m-connectivity.

Other Basic Morphological Algorithms

Thickening $A \odot B = A \cup (A \circledast B)$ ➡ In practice,

$$B = \{B^1, B^2, B^3, \cdots, B^n\}$$

$$A \odot B = A^c \otimes B$$

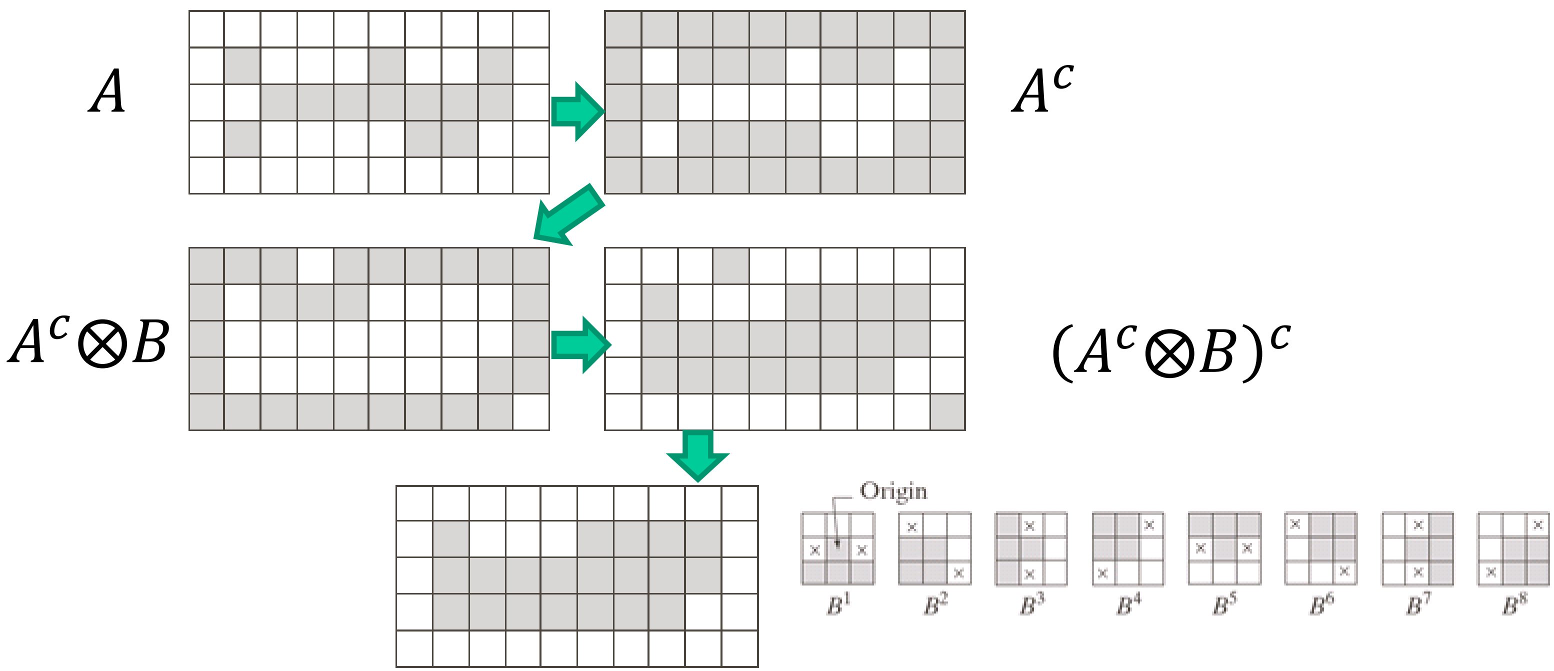


FIGURE 9.22 (a) Set A. (b) Complement of A. (c) Result of thinning the complement of A. (d) Thickened set obtained by complementing (c). (e) Final result, with no disconnected points.

Applications of Morphological Operations

- Boundary extraction
- Hole filing
- Connected component analysis
- Convex hull extraction
- Skeleton analysis

Basic Morphological Algorithms

Boundary extraction $\beta(A) = A - (A \ominus B)$

a b

FIGURE 9.14 (a) A simple binary image, with 1s represented in white. (b) Result of using Eq. (9.5-1) with the structuring element in Fig. 9.13(b).

Hole Filling

Hole: a background region surrounded by a connected foreground pixels.

Objective: given a point in a hole, fill the hole with foreground pixels.

X_0 is a set of all 0s except the selected background point
A is the set of foreground boundary

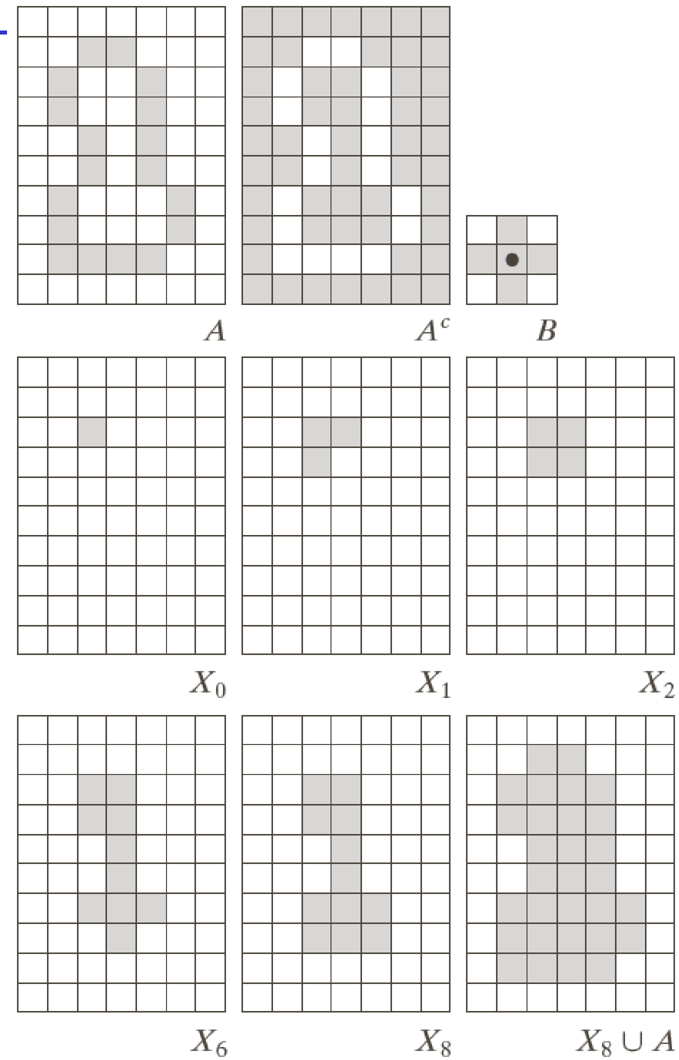
Repeat:

$$X_k = (X_{k-1} \oplus B) \cap A^c \quad \text{k=1,2,3,…}$$

Until

$$X_k = X_{k-1}$$

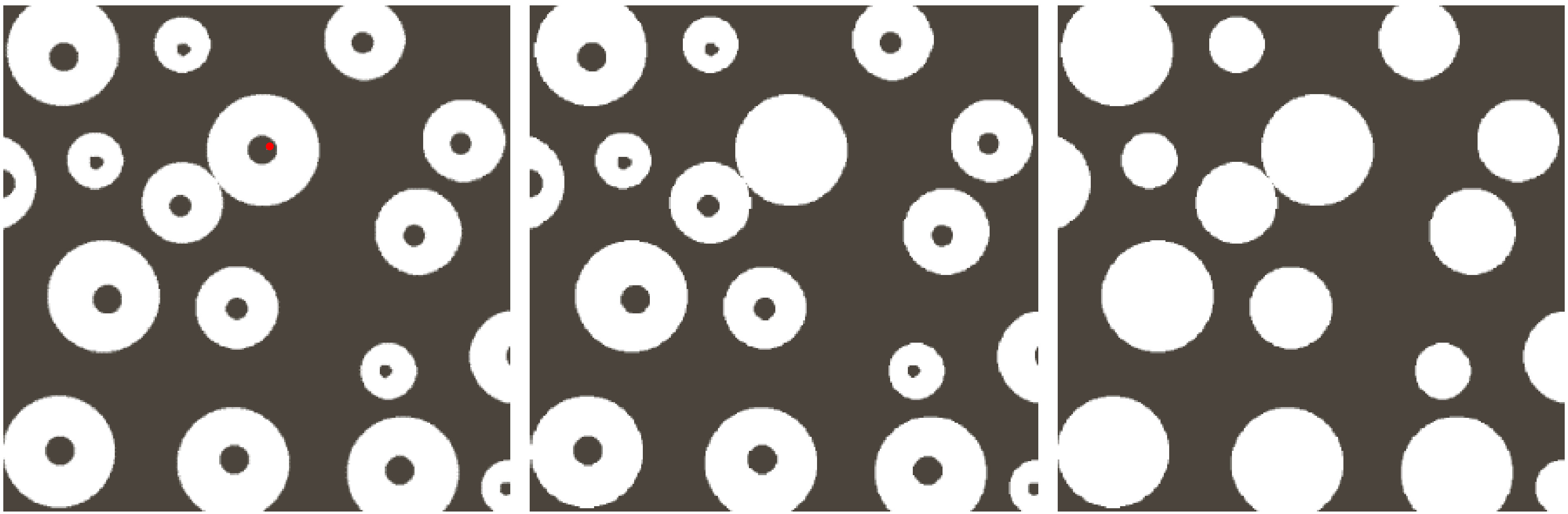
Conditional dilation



a b c
d e f
g h i

FIGURE 9.15 Hole filling. (a) Set A (shown shaded). (b) Complement of A. (c) Structuring element B. (d) Initial point inside the boundary. (e)–(h) Various steps of Eq. (9.5-2). (i) Final result [union of (a) and (h)].

Example

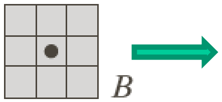


a b c

FIGURE 9.16 (a) Binary image (the red dot inside one of the regions is the starting point for the hole-filling algorithm). (b) Result of filling that region. (c) Result of filling all holes.

Connected Component Analysis

Blob extraction or region labeling



Objective: find connected components in a binary image.

Applications: finding candidates of target object for recognition

X_0 is a set of all 0s except the selected point belong to the connected component

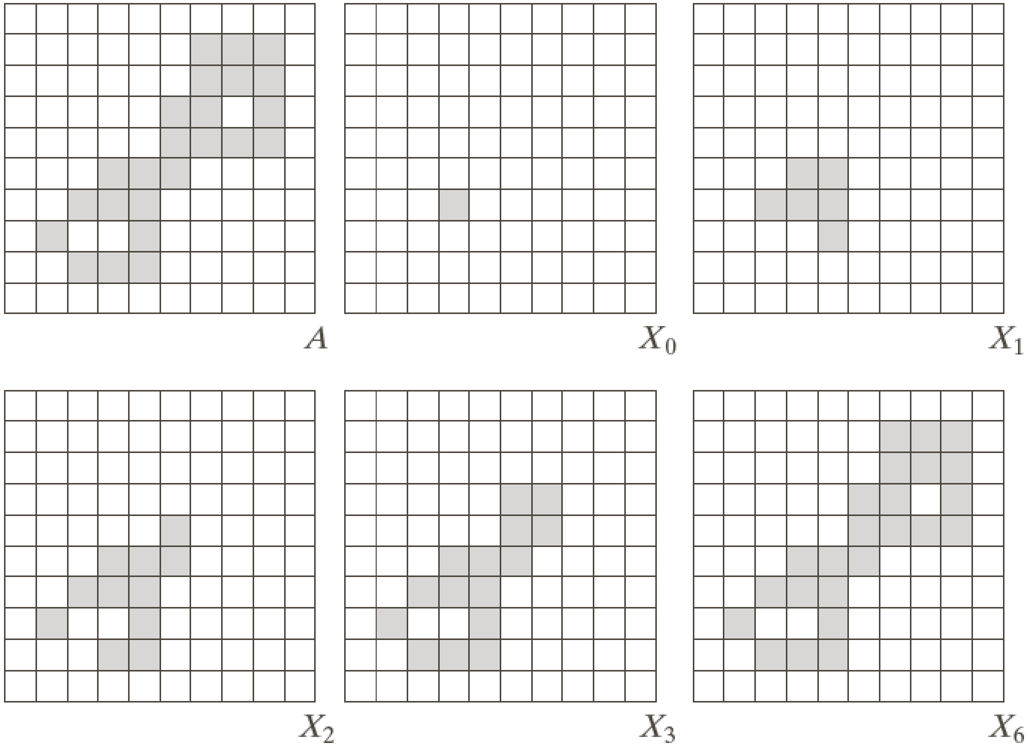
A is the set containing one or more connected components

Repeat:

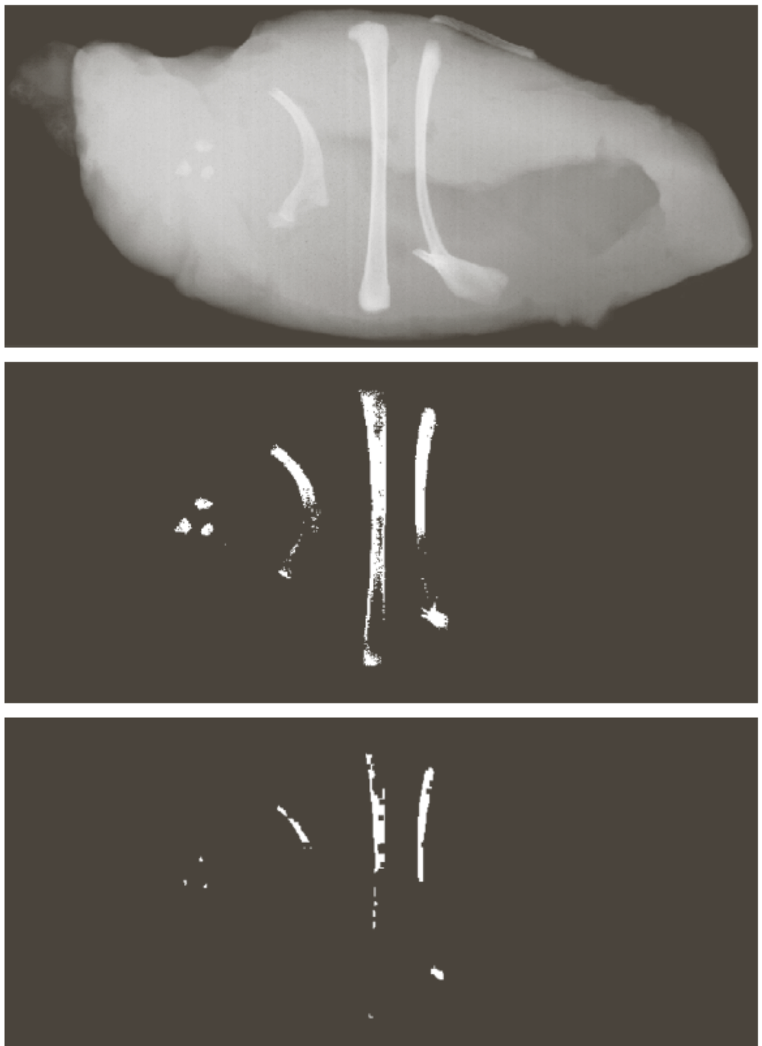
$X_k = (X_{k-1} \oplus B) \cap A$ k=1,2,3,…

Until

$X_k = X_{k-1}$



Example



Connected component	No. of pixels in connected comp
01	11
02	9
03	9
04	39
05	133
06	1
07	1
08	743
09	7
10	11
11	11
12	9
13	9
14	674
15	85

a
b
c d

FIGURE 9.18 (a) X-ray image of chicken filet with bone fragments. (b) Thresholded image. (c) Image eroded with a 5×5 structuring element of 1s. (d) Number of pixels in the connected components of (c). (Image courtesy of NTB Elektronische Geraete GmbH, Diepholz, Germany, www.ntbxray.com.)

Convex Hull

Convex: a set ***A*** is convex if the line segment connecting any two points in ***A*** is entirely belong to ***A***

Examples: rectangle, triangle, circle **are** convex

Ring, hand, many other objects with dents or hollows **are not** convex

Convex hull of a set *S*: the minimal convex set containing ***S***

Applications of finding convex hull: an abstract representation for high level image understanding

Extract Convex Hull

A is the target object and $X_0^i = A$

For each structure element $B^i, i = 1,2,3,4$

Repeat:

$$X_k^i = (X_{k-1}^i \circledast B^i) \cup A$$

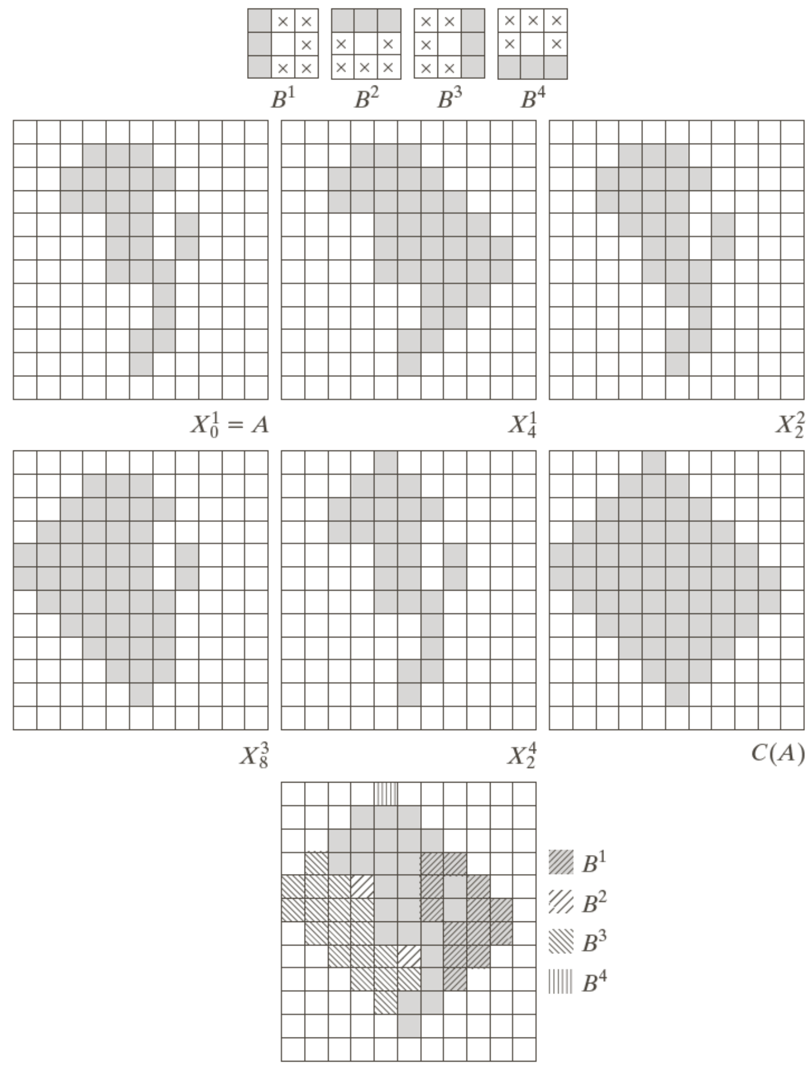
Until

$$X_k^i = X_{k-1}^i$$

Hit-or-miss

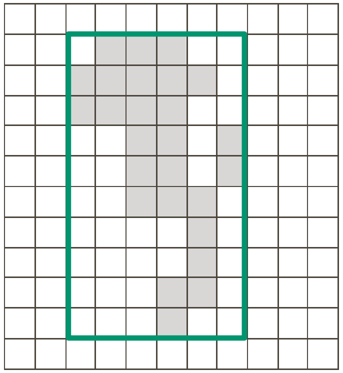
The convex hull of A is

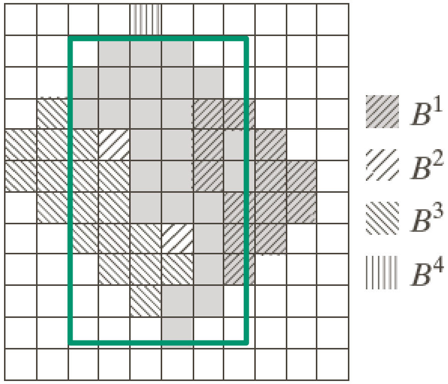
$$C(A) = \bigcup_{i=1}^{4} X_k^i$$

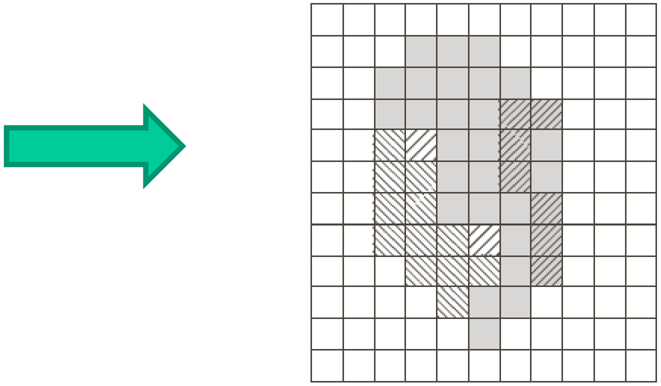


Extract Convex Hull (Cont'd)

Potential issue:

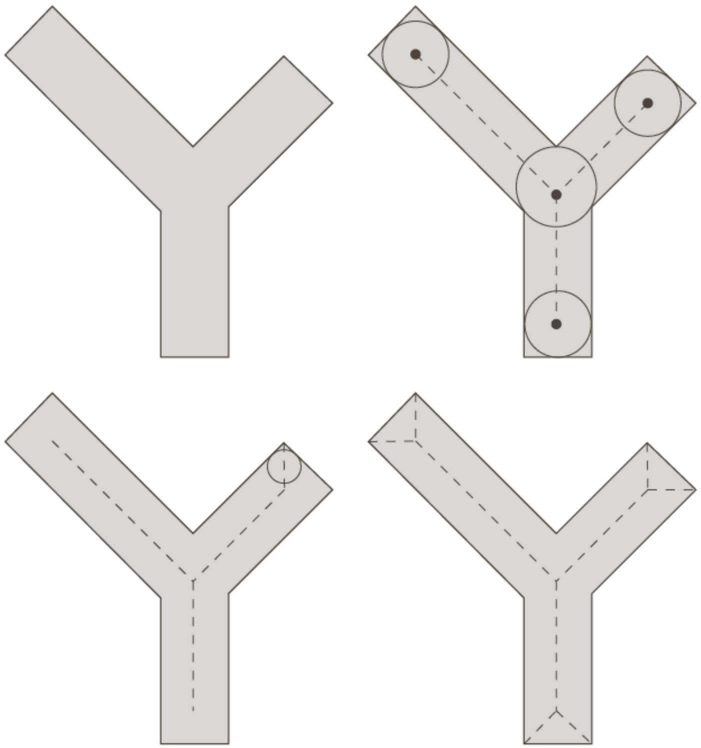






Morphological Skeleton

Skeleton (Defined by centers of maximal disks):

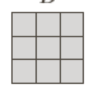


If a point Z belongs to the skeleton of A, we can find a maximal disk that entirely lies in A and touches the boundary of A at no less than two positions.

Applications: an abstract shape representation for high level image understanding, e.g. Optical Character Recognition (OCR)

Morphological Skeleton

$$S(A) = \bigcup_{k=0}^{K} S_k(A)$$



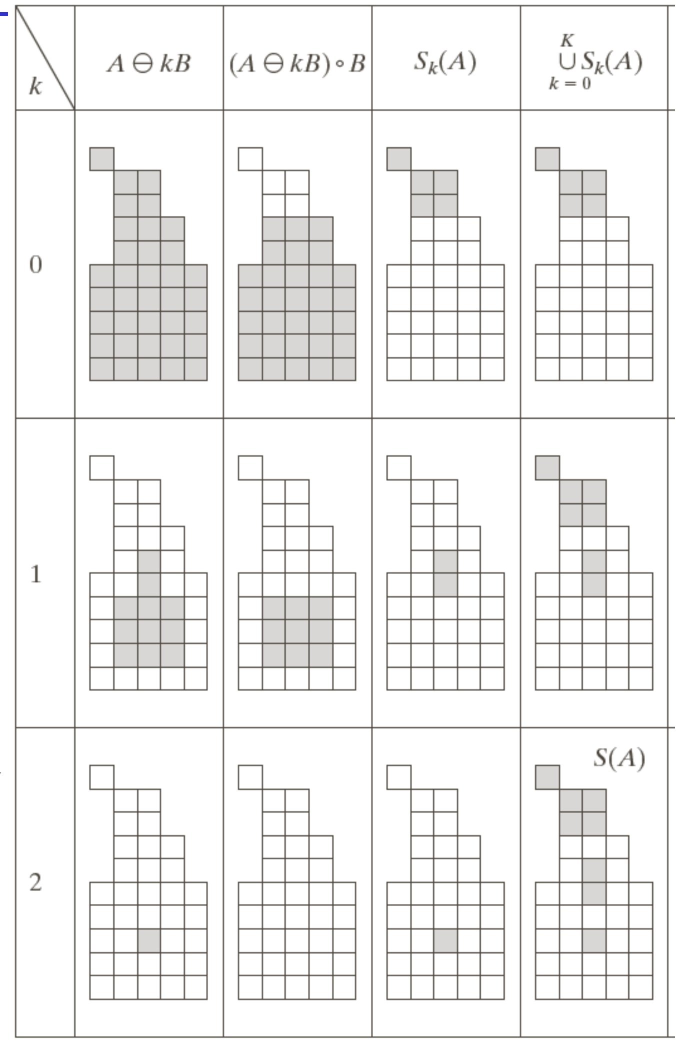
The k[th] erosion

$$S_k(A) = (A \ominus kB) - (A \ominus kB) \circ B$$

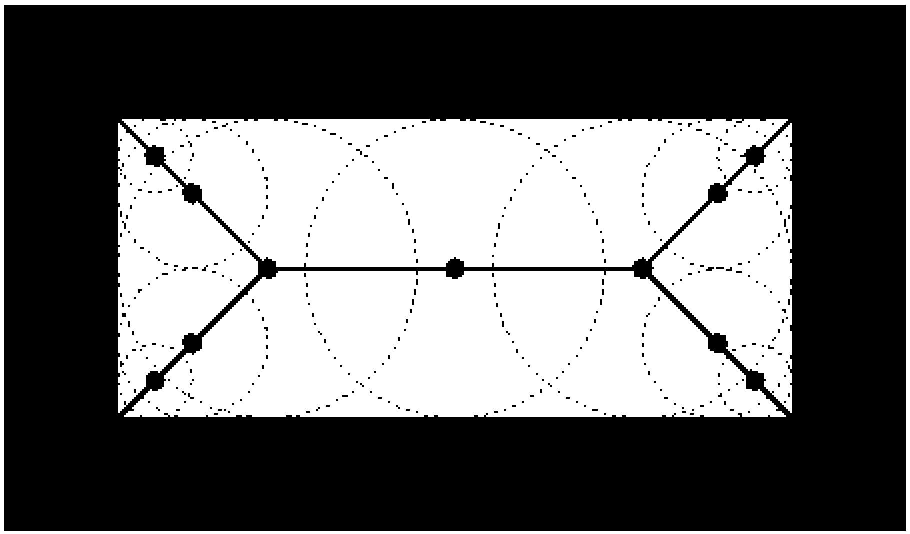
$$K = max\{k | (A \ominus kB) \neq \emptyset\}$$

Reconstruct ***A*** from its skeleton

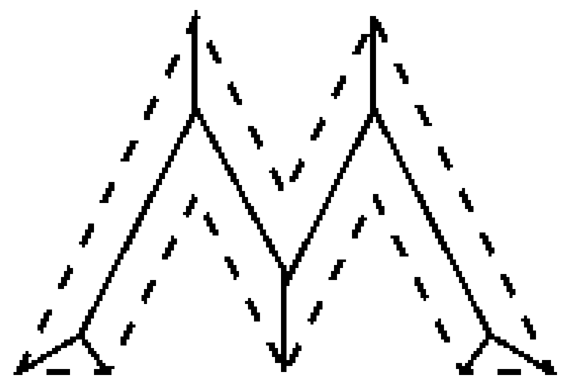
$$A = \bigcup_{k=0}^{K} (S_k(A) \oplus kB)$$



Examples

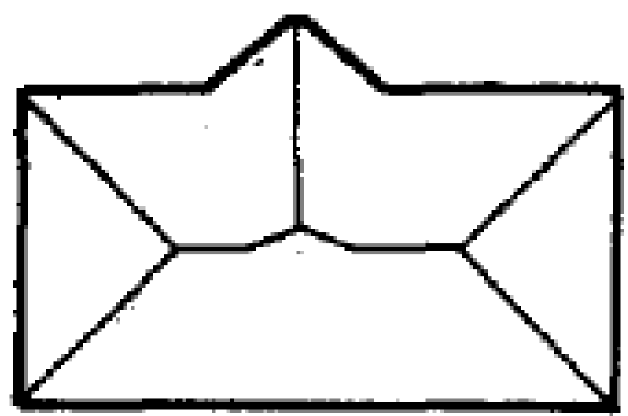


http://homepages.inf.ed.ac.uk/rbf/HIPR2/skeleton.htm



Potential issues with skeleton?

Sensitive to noise



Maragos and Schafer, “Morphological Skeleton Representation and Coding of Binary Images”, IEEE Trans. on Acoustics, Speech, and Signal Processing, Vol. 34, No. 5, 1986.